

SEQUENCES AND SUMMATIONS

Subtitle

E. Thuma and E. Zimudzi

Department of Computer Science
University of Botswana

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OUTLINE

1. INTRODUCTION TO SEQUENCES

2. RECURRENCE RELATIONS

2.1 Arithmetic Sequence

2.2 Geometric Sequence

3. SUMMATIONS

4. SUMMATION PROPERTIES AND FORMULAS

Sequence

Definition 1

Ordered list of elements created by a function to the set S .

Notation: a_n represents n^{th} term.

Example

If $a_n = 2n$, find $\{a_n\}$.

$$a_0 = 2(0) = 0$$

$$a_1 = 2(1) = 2$$

$$a_2 = 2(2) = 4$$

$$a_2 = 2(3) = 6$$

(1)

$$\therefore \{a_n\} = \{0, 2, 4, 6, \dots\}$$

A sequence can have finite or infinite terms.

Alternating Sequences

Example

Writing the first 5 terms of an alternating sequence defined by an explicit formula:

$$a_n = \frac{(-1)^n n^2}{n+1}$$

$$a_1 = -\frac{(-1)^1 1^1}{1+1} = -\frac{1}{2}$$

$$a_2 = \frac{(-1)^2 1^2}{1+1} = \frac{4}{3}$$

$$a_3 = -\frac{(-1)^3 1^3}{1+1} = -\frac{9}{4}$$

$$a_4 = \frac{(-1)^4 1^4}{1+1} = \frac{16}{5}$$

$$a_5 = -\frac{(-1)^5 1^5}{1+1} = -\frac{25}{6}$$

(2)

Factorial Notation

Example

Writing the first 5 terms of an alternating sequence defined by an explicit formula:

$$a_n = \frac{5n}{(n+2)!}.$$

$$\begin{aligned}a_1 &= \frac{5(1)}{(1+2)!} = \frac{5}{3!} = \frac{5}{6} \\a_2 &= \frac{5(1)}{(2+2)!} = \frac{10}{4!} = \frac{10}{24} = \frac{5}{12} \\a_3 &= \frac{5(1)}{(3+2)!} = \frac{15}{5!} = \frac{15}{120} = \frac{1}{8} \\a_4 &= \frac{5(1)}{(4+2)!} = \frac{20}{6!} = \frac{20}{720} = \frac{1}{36} \\a_5 &= \frac{5(1)}{(5+2)!} = \frac{25}{7!} = \frac{25}{5040} = \frac{5}{1008}\end{aligned}\tag{3}$$

Definition: Recurrence Relation

A recurrence relation for the sequence $\{a_n\}$ **equation** that expresses a_n in terms of one or more of the previous terms of the sequence, namely, $a_0, a_1, a_2, \dots, a_{n-1}$ for all integers, with $n > n_0$, where n_0 is a nonnegative integer. The **Initial conditions**, that specify the terms that precede the first term, are required to specify terms that precede the first term where the relation takes effect.

Example

$a_n = a_{n-1} + 2a_{n-2}$ for $n \geq 2$, where $a_0 = 2$, and $a_1 = 5$.

$$a_2 = a_1 + 2a_0 = 5 + 2(2) = 9$$

$$a_3 = a_2 + 2a_1 = 9 + 2(5) = 19$$

$$a_4 = a_3 + 2a_2 = 19 + 2(9) = 37$$

$$a_n = \{2, 5, 9, 19, 37, \dots\}$$

Fibonacci Sequence

The Fibonacci sequence, $f_0, f_1, f_2, \dots$, is defined by the initial conditions $f_0 = 0$, $f_1 = 1$, and the recurrence relation

$$f_n = f_{n-1} + f_{n-2}$$

for $n = 2, 3, 4, \dots$

Example Find the Fibonacci numbers f_2, f_3, f_4, f_5 , and f_6 .

Initial conditions $f_0 = 0, f_1 = 1$

$$f_2 = f_1 + f_0 = 1 + 0 = 1$$

$$f_3 = f_2 + f_1 = 1 + 1 = 2$$

$$f_4 = f_3 + f_2 = 2 + 1 = 3$$

$$f_5 = f_4 + f_3 = 3 + 2 = 5$$

$$f_6 = f_5 + f_4 = 5 + 3 = 8$$

Factorial Notation

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Solving Recurrence Relations

Finding a non-recursive formula to calculate a_n is called solving the recurrence relation. The solution is called the **closed formula**.

Example

Let a_n be the sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for $n \geq 1$ with $a_0 = 2$.

Using the method for iteration, involving substitution,

$$a_0 = 2$$

$$a_1 = 2 + 3$$

$$a_2 = (2 + 3) + 3 = 2 + 2(3)$$

$$a_3 = (2 + 3 + 3) + 3 = 2 + 3(3)$$

$$a_4 = 2 + 4(3)$$

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$$a_n = 2 + 3n$$

Example

Determine whether the sequence $\{a_n\}$, where $a_n = 3n$ for every nonnegative integer n , is a solution for a recurrence relation $a_n = 2a_{n-1} - a_{n-2}$ for $2, 3, 4, \dots$.

Solution

Suppose $a_n = 3n$ for every nonnegative integer n . Then, for $n \geq 2$, we see that $2a_{n-1} - a_{n-2} = 2(3(n-1)) - 3(n-2) = 3n = a_n$. Therefore, $\{a_n\}$, where $a_n = 3n$ is a solution to the recurrence relation.

String Sequences

Finite sequences are also called strings.

The length of the string is the number of terms in the string. The empty string λ has no terms.

e.g. *abcd* is a string of length 4.

Some Useful Sequences.	
<i>nth Term</i>	<i>First 10 Terms</i>
n^2	1, 4, 9, 16, 25, 36, 49, 64, 81, 100, ...
n^3	1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, ...
n^4	1, 16, 81, 256, 625, 1296, 2401, 4096, 6561, 10000, ...
f_n	1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, ...
2^n	2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, ...
3^n	3, 9, 27, 81, 243, 729, 2187, 6561, 19683, 59049, ...
$n!$	1, 2, 6, 24, 120, 720, 5040, 40320, 362880, 3628800, ...

Polynomial Fitting

We say that a sequence is Δ^k -constant if the k^{th} differences are constant.

Example

For the sequence 2, 3, 7, 14, 24, 37, $\dots$,

First differences are 1, 4, 7, 10, 13

Second differences are 3, 3, 3, 3, 3

So the sequence is Δ^2 -constant, so the formula for a_n will be degree 2 polynomial, i.e.

we know for some constants a , b , and c that

$$a_n = an^2 + bn + c$$

$a_0 = 2$ from the sequence

Now plug in $n = 1$ and $n = 2$, we get $a_1 = 3$ and $a_2 = 7$

i.e. $3 = a + b + 2$

and $7 = 4a + 2b + 2$

Solving, we find $a = \frac{3}{2}$ and $b = -\frac{1}{2}$

So, $a_n = \frac{3}{2}n^2 - \frac{1}{2}n + 2$

Example

Use polynomial fitting to find the formula for the n^{th} term of the sequence $(a_n), n \geq 0$ below.

We say that a sequence is Δ^k -constant if the k^{th} differences are constant.

Example

a. $0, 2, 6, 12, 20, \dots$

First differences are $2, 4, 6, 8$

Second differences are $2, 2, 2, 2$

So, $an^2 + bn + c$

Since $a_0 = 0$, we know that $c = 0$.

We plug in 2 (a_1) and 6 (a_2) from the sequence, i.e. 2^{nd} and 3^{rd} terms respectively.

We solve the simultaneous equations

$$2 = a + b$$

$$6 = 4a + 2b$$

Solving the simultaneous equations we get $a = 1$, and $b = 1$

Finding an Explicit Formula

The key to finding an explicit formula when given several terms of the sequence is to look for the pattern in the terms. The pattern may include alternating terms, formula for denominators, formula for numerators, exponents and bases.

Procedure

- Look for the pattern among the terms.
- If the terms are fractions, look for a separate pattern among numerators and denominators.
- Look for a pattern among the signs of the terms.
- Write a formula for a_n in terms of n . Test your formula for $n = 1$, $n = 2$, and $n = 3$.

Example Find formulae/rule for the sequences with the following first five terms:

(a) $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}$

(b) $1, 3, 5, 7, 9$

(c) $1, -1, 1, -1, 1$.

Solution:

(a) We recognize that the denominators are powers of 2. The sequence with $a_n = \frac{1}{2^n}$, $n = 0, 1, 2, \dots$ is a possible match. This proposed sequence is a geometric progression with $a = 1$ and $r = \frac{1}{2}$.

(b) We note that each term is obtained by adding 2 to the previous term. The sequence with $a_n = 2n + 1$, $n = 0, 1, 2, \dots$ is a possible match. This proposed sequence is an arithmetic progression with $a = 1$ and $d = 2$.

(c) The terms alternate between 1 and -1. The sequence with $a_n = (-1)^n$, $n = 0, 1, 2, \dots$ is a possible match. This proposed sequence is a geometric progression with $a = 1$ and $r = -1$.

Example Write an explicit formula for the n^{th} of each sequence:

(a) $a_n = \left\{-\frac{2}{11}, \frac{3}{13}, -\frac{4}{15}, \frac{5}{17}, -\frac{6}{19}, \dots\right\}$

(b) $a_n = \left\{-\frac{2}{25}, -\frac{2}{125}, -\frac{2}{625}, -\frac{2}{3125}, -\frac{2}{15625}, \dots\right\}$

(c) $a_n = \left\{-\frac{3}{4}, -\frac{9}{8}, -\frac{27}{12}, -\frac{81}{16}, -\frac{243}{20}, \dots\right\}$

Solution:

(a) The terms alternate between positive and negative. We can use $(-1)^n$ to make the terms alternate. The numerator can be represented by $n + 1$. The denominator can be represented by $2n + 9$.

Therefore, $a_n = \frac{(-1)^n(n+1)}{2n+9}$.

(b) The numerator is 2. The denominator are increasing powers of 5, i.e. $\left\{-\frac{2}{5^2}, -\frac{2}{5^3}, -\frac{2}{5^4}, -\frac{2}{5^5}, \dots\right\}$.

The fraction is negative, the numerator is 2, and the denominator can be represented as 5^{n+1} .

Therefore, $a_n = -\frac{2}{5^{n+1}}$

(c) The fraction is negative. The numerator are powers of 3, the denominators are multiples of 4. Therefore, $a_n = -\frac{3^n}{4n}$

Definition: Arithmetic Sequence

A sequence formed by adding the initial term, a , and the product of the common difference, d , and the term number, n .

$$\{a_n\} = a, a + d, a + 2d, a + 3d, \dots, a + nd \dots$$

The *initial term* a and the *common difference* d are real numbers.

Remark: An arithmetic progression is a discrete analogue of the linear function $f(x) = dx + a$.

Example

Let $a = 5$ and $d = 2$. Find the first 5 terms of $\{a_n\}$.

Sol $a_n = 5 + 2n$

Example

$\{s_n\}$ where $s_n = -1 + 4n$ is an arithmetic progression with initial term -1 and common difference 4.

Example

Find a and d for the sequence $\{7, 4, 1, -2, \dots\}$

Definition: Geometric Sequence

A sequence formed by multiplying the initial term, a , by the common ratio to the n^{th} power, r .

The *initial term* a and the *common ratio* r are real numbers.

$$\{a_n\} = a, ar, ar^2, ar^3, \dots, ar^n$$

Remark: A geometric progression is a discrete analogue of the exponential function $f(x) = ar^x$.

Example

Let $a = 4$ and $r = 3$. Find the first 5 terms of the geometric progression $\{a_n\}$.

Sol $a_n = 4(3)^n$, and $\{a_n\} = \{4, 12, 36, 108, 324, \dots\}$

Example

Find a and r for the geometric sequence $\{a_n\} = \{3, \frac{3}{2}, \frac{3}{4}, \frac{3}{8}, \dots\}$.

$$a_n = ar^n = 3r^n, r = \frac{1}{2}$$

SUMMATIONS

Sigma Notation

if m and n are integers and $m \leq n$, then, the 'summation from k equals m to n of a_k is the sum of all the terms $\{a_m, a_{m+1}, \dots, a_n\}$. We write

$$\sum_{k=m}^n a_k = \{a_m + a_{m+1} + a_{m+2} \cdots + a_n\}$$

and call k the **index** of the summation, m the **lower limit** of the summation, and n the **upper limit** of the summation.

Example

Use sigma notation to express the sum of the first 100 terms of the sequence a_i where $a_i = \frac{1}{i}$ for $i = 1, 2, \dots$

$$a_1 = \frac{1}{1}, a_2 = \frac{1}{2}, a_3 = \frac{1}{3} \cdots, a_i = \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{100}$$

$$\sum_{i=1}^{100} \frac{1}{i} = \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{100}$$

Example What is the value of $\sum_{i=5}^9 i^2$?

Solution

$$\begin{aligned}\sum_{i=5}^9 i^2 &= 5^2 + 6^2 + 7^2 + 8^2 + 9^2 \\ &= 25 + 36 + 49 + 64 + 81 \\ &= 255\end{aligned}$$

Example What is the value of $\sum_{i=7}^{10} (-1)^i$?

Solution

$$\begin{aligned}\sum_{i=7}^{10} (-1)^i &= (-1)^7 + (-1)^8 + (-1)^9 + (-1)^{10} \\ &= -1 + 1 - 1 + 1 \\ &= 0\end{aligned}$$

Example Find the following sum:

$$\sum_{n=0}^4 \frac{n!}{2}$$

Solution

$$\begin{aligned}\sum_{n=0}^4 \frac{n!}{2} &= \frac{0!}{2} + \frac{1!}{2} + \frac{2!}{2} + \frac{3!}{2} + \frac{4!}{2} \\ &= \frac{1}{2} + \frac{1}{2} + \frac{2}{2} + \frac{3 \cdot 2}{2} + \frac{4 \cdot 3 \cdot 2}{2} \\ &= \frac{1}{2} + \frac{1}{2} + 1 + 3 + 12 \\ &= 17\end{aligned}$$

Example Write the following sums using summation notation:

a. $1 + 3 + 5 + \cdots 17$

b. $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} \cdots + \frac{1}{117}$

Solution

a. The terms of the sum 1, 2, 3, etc., form an arithmetic sequence with first term $a = 1$ and common difference $d = 2$. $a_n = 1 + (n - 1)d = 2n - 1, n \geq 1$. To determine the upper limit of the summation, we need to determine which values of n produces the term 117, i.e.

if $a_n = 117$,

then $2n - 1 = 117$.

Therefore,

$$1 + 3 + 5 + \cdots + 117 = \sum_{n=1}^{59} (2n - 1)$$

b. We rewrite all the terms as fractions, i.e.

$$\frac{1}{1} + \frac{-1}{2} + \frac{1}{3} + \frac{-1}{4} \cdots + \frac{1}{117}$$

The numerators can be described by the geometric sequence $(-1)^{n-1}$ for $n \geq 1$, while the denominators are given by the arithmetic sequence n for $n \geq 1$. The upper limit is 117.

Thus,

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} \cdots + \frac{1}{117} = \sum_{n=1}^{117} \frac{(-1)^{n-1}}{n}$$

Change of index

Suppose we have the sum

$$\sum_{j=1}^5 j^2$$

but want the index of summation to run between 0 and 4 rather than from 1 to 5. To do this, we let $k = j - 1$. Then the new summation index runs from 0 (because $k = 1 - 1 = 0$ when $j = 1$) to 4 (because $k = 5 - 1 = 4$ when $j = 5$), and the term j^2 becomes $(k + 1)^2$. Hence,

$$\sum_{j=1}^5 j^2 = \sum_{k=0}^4 (k + 1)^2$$

Separating Off the Last Term

Example

Rewrite

$$\sum_{t=1}^{n+1} \frac{1}{t^2}$$

by separating off the final term.

Solution

$$\sum_{t=1}^{n+1} \frac{1}{t^2} = \sum_{t=1}^n \frac{1}{t^2} + \frac{1}{(n+1)^2}$$

Example

Write $\sum_{k=0}^n t^k + 2^{n+1}$ as a single summation.

Solution

$$\sum_{k=0}^n t^k + 2^{k+1} = (2^0 + 2^1 + 2^2 + \dots + 2^n) + 2^{n+1} = \sum_{k=0}^{n+1} t^k$$

Basic Properties of Summations

$$\sum_{k=1}^n ca_k = c \sum_{k=1}^n a_k$$

$$\sum_{k=1}^n c = cn$$

$$\sum_{k=1}^n (a_k \pm b_k) = \sum_{k=1}^n a_k \pm \sum_{k=1}^n b_k$$

$$\sum_{k=1}^n a_k = \sum_{k=1}^j a_k + \sum_{k=j+1}^n a_k$$

$$\sum_{k=1}^n a_k = \sum_{k=1+m}^{n+m} a_{k-m}$$

for $i \leq j < j+1 < n$

Useful Summation Formula

Some summations arise all the time in discrete mathematics. For such summations, it is often useful to derive a closed form, i.e. without summations.

$$\sum_{i=1}^n i = \frac{(n)(n+1)}{2}$$

Example $\sum_{i=1}^7 i = \frac{(7)(8)}{2} = 28$

$$\sum_{i=1}^n i^2 = \frac{(n)(n+1)(2n+1)}{6}$$

Example $\sum_{i=1}^7 i^2 = \frac{(7)(8)(2(7)+1)}{6} = 140$

$$\sum_{i=1}^n i^3 = \frac{(n^2)(n+1)^2}{4}$$

Example $\sum_{i=1}^7 i^3 = \frac{(7^2)(8^2)}{4} = 784$

$$\sum_{k=0}^n ar^k = \frac{(ar^{n+1} - 1)}{r - 1}, r \neq 1$$

Example $\sum_{k=0}^7 (2.3^k + 5.2^k) = \sum_{k=0}^7 2.3^k + \sum_{k=0}^7 5.2^k$

Example

$$\begin{aligned} \sum_{k=0}^7 (2.3^k + 5.2^k) &= \sum_{k=0}^7 2.3^k + \sum_{k=0}^7 5.2^k \\ &= \frac{2.3^8 - 2}{2} + \frac{5.2^8 - 5}{1} \\ &= \frac{13122 - 2}{2} + \frac{1275}{1} \\ &= 7835 \end{aligned}$$

Example - Using known formulae

Find $\sum_{k=50}^{100} k^2$

Solution

Note that $\sum_{k=1}^{100} k^2 = \sum_{k=1}^{49} k^2 + \sum_{k=50}^{100} k^2$, we have

$$\sum_{k=50}^{100} k^2 = \sum_{k=1}^{100} k^2 - \sum_{k=1}^{49} k^2,$$

Using the formula $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$

$$\sum_{k=50}^{100} = \frac{100(101)(201)}{6} - \frac{49(50)(99)}{6} = 297925$$

Example - Double Summations

To evaluate the double sum $\sum_{i=1}^4 \sum_{j=1}^3 ij$, first expand the inner summation and then continue by computing the outer summation.

$$\sum_{i=1}^4 \sum_{j=1}^3 ij = \sum_{j=1}^4 (i + 2i + 3i) = \sum_{j=1}^4 6i = 60$$

The End